9. PLACENTA AND OTHER EMBRYONIC ANNEXES

DEFINITION

Embryonic annexes (or fetal annexes) are structures that form in parallel during the development of the embryo and then the fetus. They ensure the vital functions of respiration, nutrition, and excretion. These embryonic annexes, which are located between the fetus and the mother's uterus, are the placenta (which ensures the exchange of substances between the mother and the fetus), the amnion (the membrane delimiting the amniotic cavity containing the amniotic fluid and which lines the inner wall of the placenta), the allantois (which participates in the formation of the placenta), and the yolk sac (which produces the first sex cells and red blood cells). In chronological order, the following appeared:

- of the amnion and yolk sac (8 days post-fertilization),
- of the placenta (9 days = lacunar stage of the syncytiotrophoblast)
- allantoin (16 days).

1. PLACENTA

1.1. Definition and characteristics

A transient organ resulting from the association of the chorion of the egg with the endometrium. Expelled 15 to 30 minutes after birth, it is a disc with irregular edges, 20cm in diameter, 3cm thick, weighing 500g: 1/6th of the newborn's weight.The placenta, the site of physiological exchanges between the mother and the fetus, is a "mixed" embryonic appendage made up of:

- of extra-embryonic tissues (trophoblast, extra-embryonic mesenchyme, blood vessels)
- of maternal tissues (decidua, maternal blood)

The human placenta is:

- discoidal: disc-shaped.
- pseudocotyledonate: the placental villi are grouped in clusters separated by incomplete septa.
- decidual: its expulsion results in the loss of part of the uterine lining.
- hemochorial: the trophoblast comes into contact with maternal blood.
- chorioallantoid: the placental (chorionic) circulation is connected to the fetal circulation via the allantois.

It has two sides:

- **fetal face**: shiny and smooth, in the center is the umbilical cord.
- **maternal face**: very hemorrhagic, formed of 10 to 40 irregularly polygonal zones: the placental cotyledons.

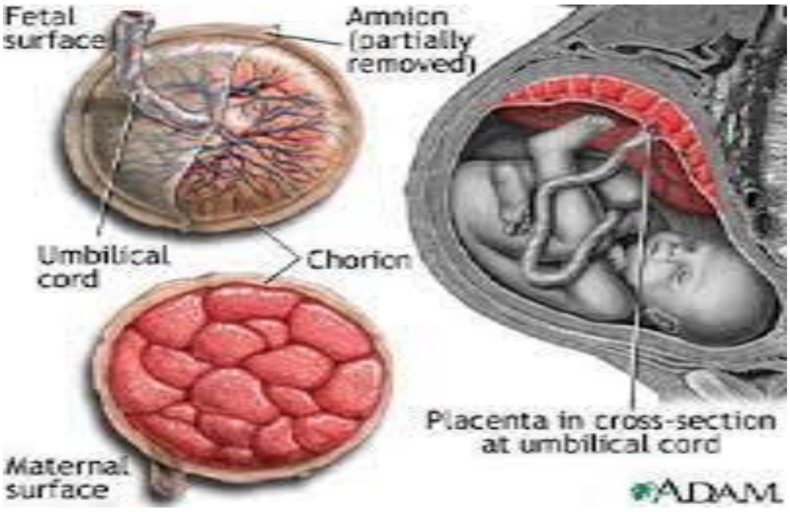


Figure1.Sides of the placenta

1.2. Organization

During placental formation, there are 6 phases:

a. Chorion avillous: when the syncytiotrophoblast is located at one of the poles of the egg.

b- Chorion avillous, fruste lacunaire: when the syncytiotrophoblast covers the surface of the egg and erodes the maternal capillaries, maternal blood flows into the lacunae.

c- Trabecular chorion: when the villi are primary.

d- Diffuse villous chorion: when secondary and tertiary villi form all over the surface of the egg.

e- Diffuse chorion, bushy chorion: uneven distribution of tertiary villi.

f- Placenta and smooth chorion: grouping of villi on one side, forming the placenta.

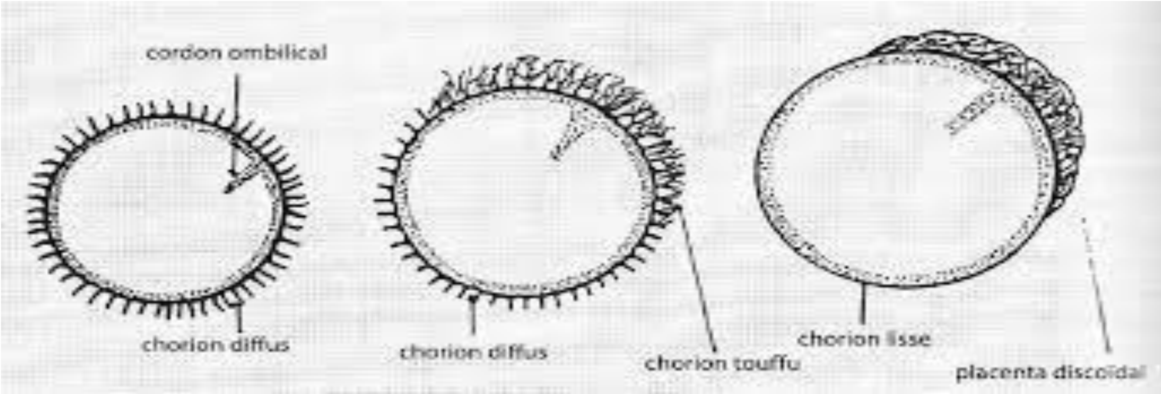


Figure2. Organization of placental villi

1.3. Structure

From the 13th day of development, the villi begin to develop, initially composed solely of syncytiotrophoblast, then cytotrophoblast penetrates these young villi, resulting in mature primary villi.

The syncytial columns progress and open maternal vessels: maternal blood spreads into the lacunae: placental maternal circulation is established.

Then, towards the end of the 3rd week, the mesenchyme of the chorionic plate begins to develop, and we will have secondary villi.

Next, vascular primordia develop inside the villi, forming the tertiary villi.

These vascular primordia enter the pedicle of fixation and then the mesenchyme of the villi: The future fetal circulation begins.

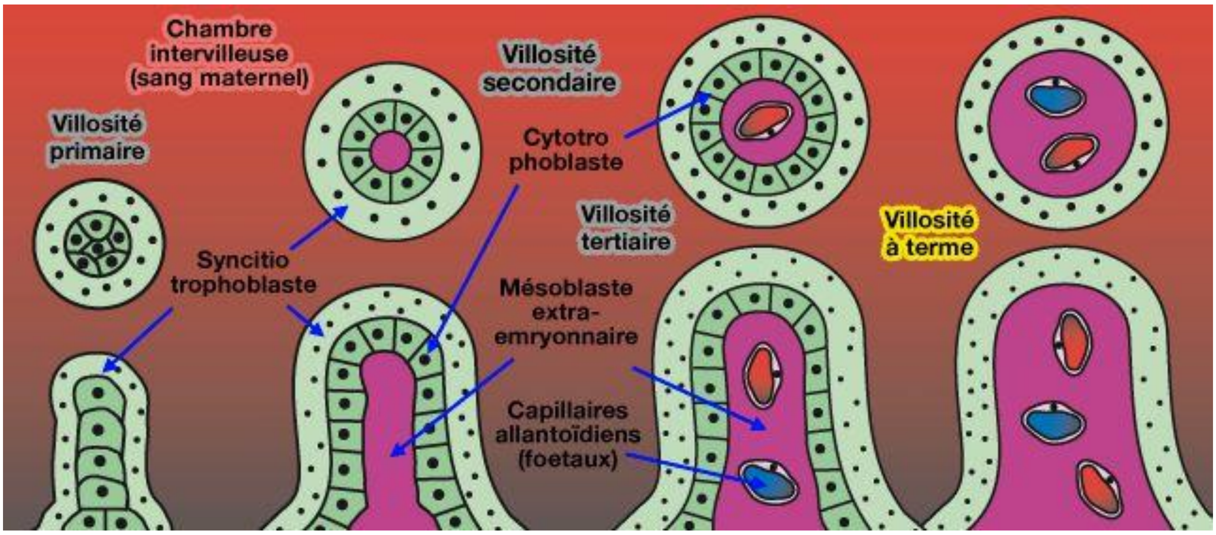


Figure 3. Placental villi

Around the 21st day, the intravillous vascular network connects to the umbilical-allantoid vessels: the placental fetal circulation is established.

From the 2nd to the 4th month, the villus becomes arborized: arboriform villi, some branches attach to the maternal tissue: clinging villi, others remain free: floating villi.

The chorionic villi contain fetal blood in the intravillous space; these villi are bathed in intervillous spaces or chambers filled with maternal blood, so the fetal blood never mixes with the maternal blood. The collection of clinging and free villi originating from the same trunk constitutes a villous tree.

There are 20 to 30 large villous trunks that correspond to the cotyledons; each trunk floats in a chamber partitioned by intercotyledonary septa: partitions that appear at 4th months starting from the maternal side but not reaching the fetal side: pseudo-cotyledonous placenta.

The general structure of the placenta is established by the 5th month and is characterized by:

- ***The fetal placenta:*** consisting of the chorionic plate (amnion + ST = CT + MES), the intervillous space or zone of blood lakes where the placental villi bathe.
- ***The maternal placenta*****:** consisting of the basal plate (ST + CT + basilar decidua).
- Fetal blood draws the elements it needs from the maternal blood through the walls of the villi.

The cytotrophoblast disappears from the placental barrier from the 4th month onwards, thus reducing the distance between fetal and maternal vessels.

1.4. Physiology and functional properties

- **Selective filter**

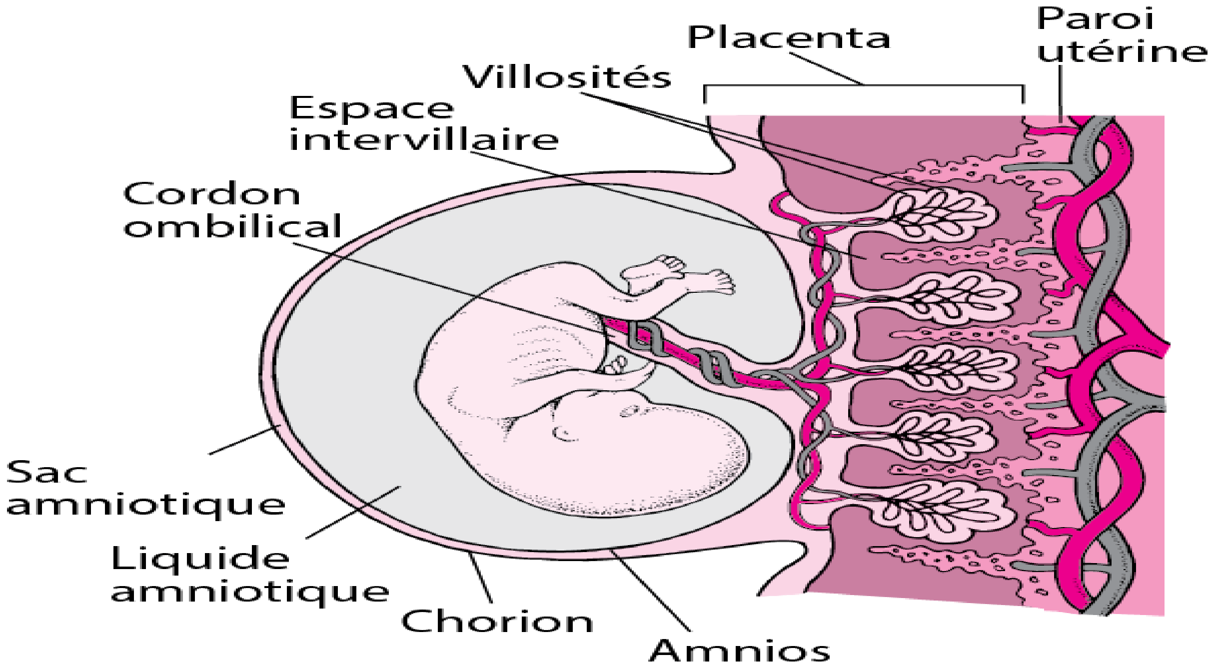
The placenta ensures the transfer of molecules between the maternal and fetal compartments. And it selects molecules according to their size, electrical charge, and spatial configuration. This filter is imperfect: bacteria, viruses, parasites, and fetal cells can pass through it.

- **Exchange properties**

The placenta is the site of fetomaternal exchanges;

- from maternal blood to fetal blood: exchanges of O2, water and electrolytes, carbohydrates, lipids and proteins, vitamins.
- The water passes through osmosis

- The mother's proteins are broken down into acids and transported by an active mechanism. Antibodies cross the barrier via pinocytosis, except for high molecular weight IgM antibodies, which do not.
- Carbohydrates pass through facilitated diffusion. Lipids are broken down and then rebuilt in the placenta.



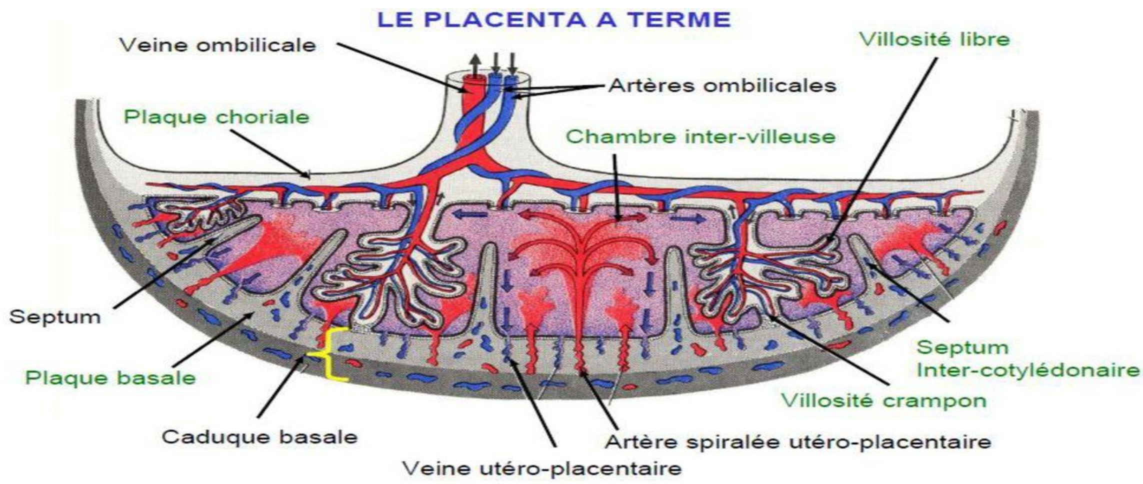


Figure4. Structures of the placenta at term

- **Endocrine function**

The placenta is a large endocrine gland. Numerous steroid and protein hormones are produced there, primarily by the trophoblast. These hormones are necessary for maintaining the pregnancy and for fetal growth.The placenta produces and releases steroid hormones and glycoproteins:

- Progesterone: plays a role in maintaining the myometrium at rest during pregnancy.
- Estrogens: play a role in the development of the mammary gland and in milk production.
- Human chorionic gonadotropin (hCG): glycoprotein hormone whose function is to transform the cyclic corpus luteum into the corpus luteum of pregnancy and to stimulate the synthesis of progesterone.
- Placental lactogen hormone (hPL) : synthesized by the syncytiotrophoblast and secreted into the maternal circulation around the 5th week, its role is to prepare the mammary gland for lactation. It also acts on fetal growth.

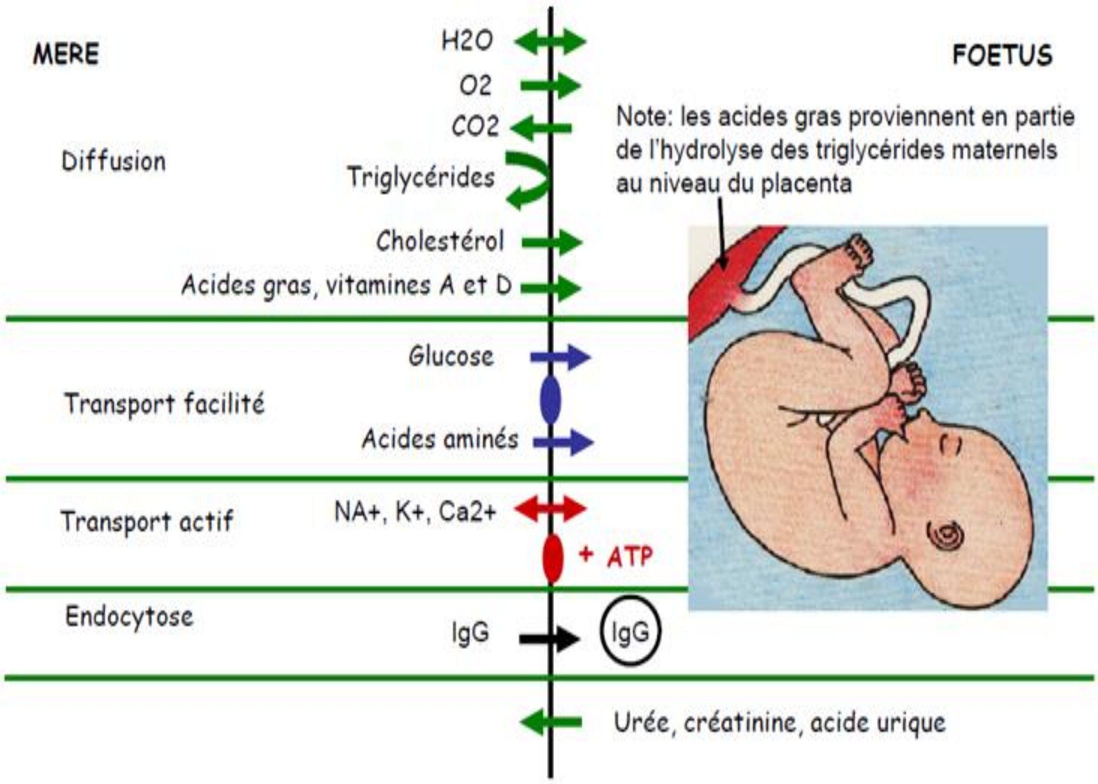


Figure 5. Metabolic exchanges across the placenta

Immunological barrier function

The placenta is located at the interface of two immune systems. The fetus is a semi-allogeneic graft. Fetal cells display histocompatibility proteins, or HLA (Human Leukocyte Antigen) proteins, on their surfaces that differ from those of the mother. For the mother's immune system, the fetus constitutes a "non-self," yet it is tolerated. The trophoblast of the villi does not express histocompatibility proteins. Furthermore, the placenta blocks the effects of maternal cytotoxic T lymphocytes through the secretion of several paracrine factors.

1.5. Placental Pathologies

Most often, the embryo implants in the upper part of the dorsal surface of the uterus.

- **Placenta previa**

Ectopic implantation at the internal cervical os uterine, can cause bleeding during pregnancy. Natural childbirth is impossi

- **Retroplacental hematoma**

Premature placental abruption normally inserted in the 3rd quarter

- **Placenta accreta**

This is an insertion of theplacentain the myometrium.

Three types are distinguished according to the depth of insertion. Among the villi in the myometrium, we can distinguish:

- **the placenta accreta itself:**t he villi are in contact with the myometrium.
- **placenta increta:** the villi invade the myometrium**.**
- **Placenta percreta:** the villi extend beyond the myometrium, sometimes invading neighboring organs (bladder).

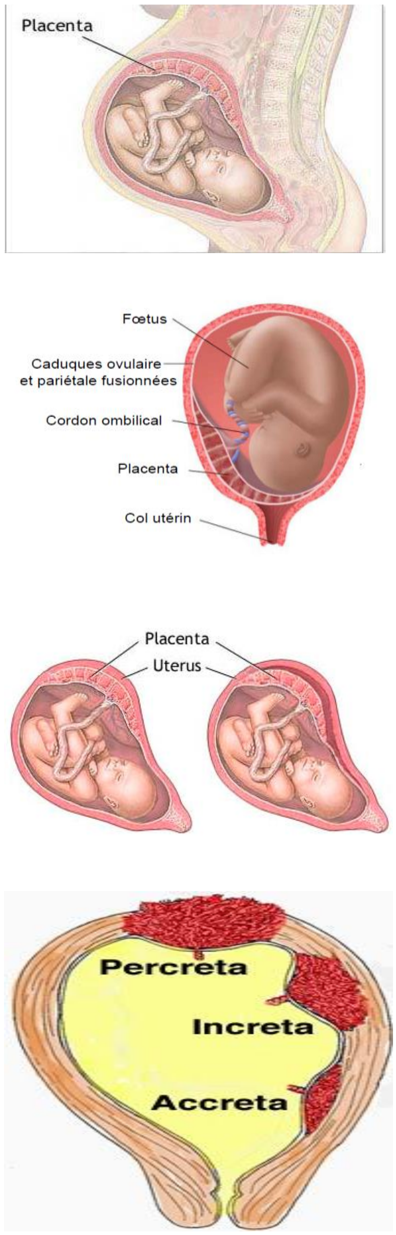


Figure 6. Placental pathologies

2. OTHER EMBRYONIC ANNEXES

In the case of mammals and the human species, these appendages are, in order of appearance: The amnion or amniotic cavity, the yolk sac, the allantois, and the umbilical cord. Each of these plays a particular role in development.

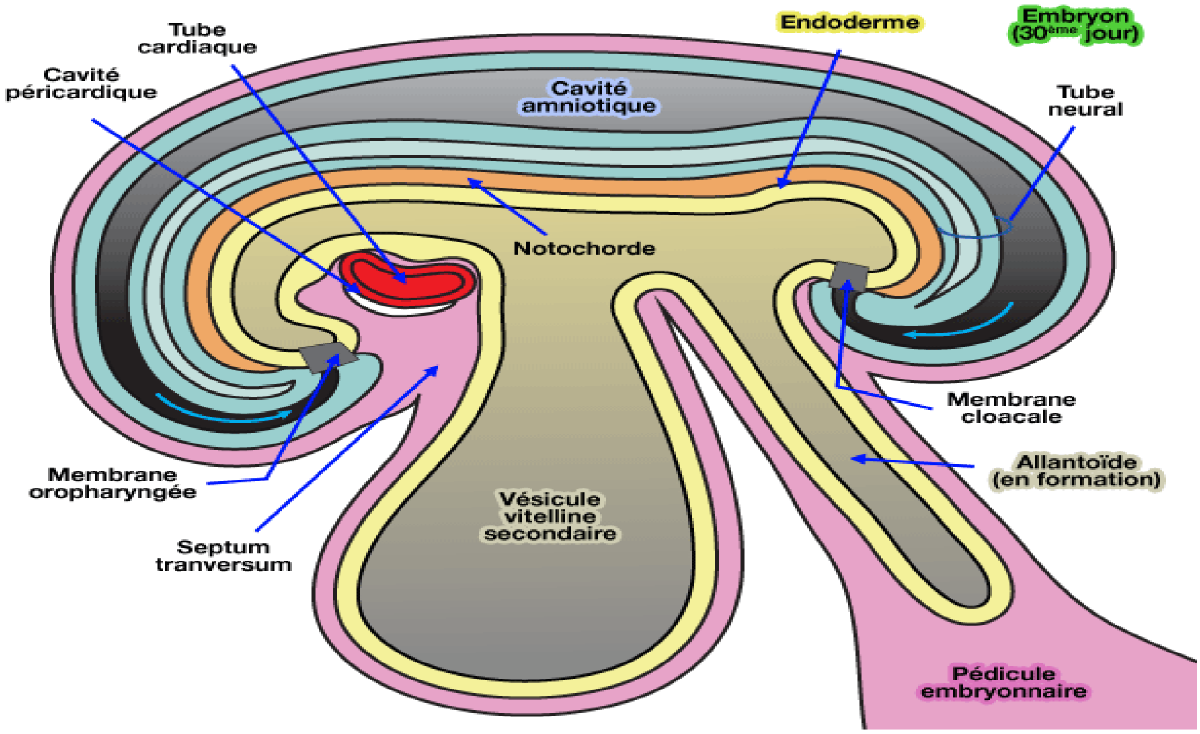


Figure 7. Embryonic annexes

2.1. Amniotic cavity and amnion

Around the 8th day of gestation, the ectoderm develops a small cavity which, as it enlarges, forms the amniotic cavity. Its roof is lined by the amnioblasts that make up the amnion, and its floor is formed by the ectoderm, which is continuous with the amnion.

- The amnion is a sac that surrounds the embryo and later the fetus. It appears on the 8th day from the cytotrophoblast and it associates with the extra-embryonic mesenchyme when the external coelom becomes distinct. The two formations, closely joined, constitute the amniotic plate (somatopleure). At the end of the 4th week, once the delimitation is complete, the amnion completely envelops the embryo.

- Amniotic fluidis a very dilute solution (99% water) of electrolytes, glucose, proteins with bactericidal properties and lipids: it contains suspended amnion cells and fetal epidermal cells.

This fluid is initially secreted by the cells of the amnion. It results from transudation from the mother's blood. During the second half of pregnancy, it is primarily produced by the fetal kidneys. It results from significant water exchange, renewed every 3 hours, and is reabsorbed by the fetus at the end of pregnancy at a rate of 400 ml/day.

The embryo, and later the fetus, floats freely in the amniotic fluid which allows symmetrical external growth, the evolution of fetal movements and what promotes the osteo-muscular organization and that of the joints. This provides mechanical protection and thermal regulation.

Amniocentesis allows the collection of cells suspended in the amniotic fluid, their culture to perform karyotyping and establish prenatal diagnosis of chromosomal abnormalities.

The amount of fluid in the amnion varies with gestational age. Abnormalities in amniotic fluid volume are detected by ultrasound:

- in the long term: on average 1 litre (0.5 and 2 litres).

- beyond 2 liters: polyhydramnioswhich refers to a sometimes considerable increase in the amount of amniotic fluid, often symptomatic of atresia or agenesis of the fetal esophagus.

- below 0.5 liters: oligohydramnios which occurs in case of rupture of the amnion with chronic leakage of fluid.

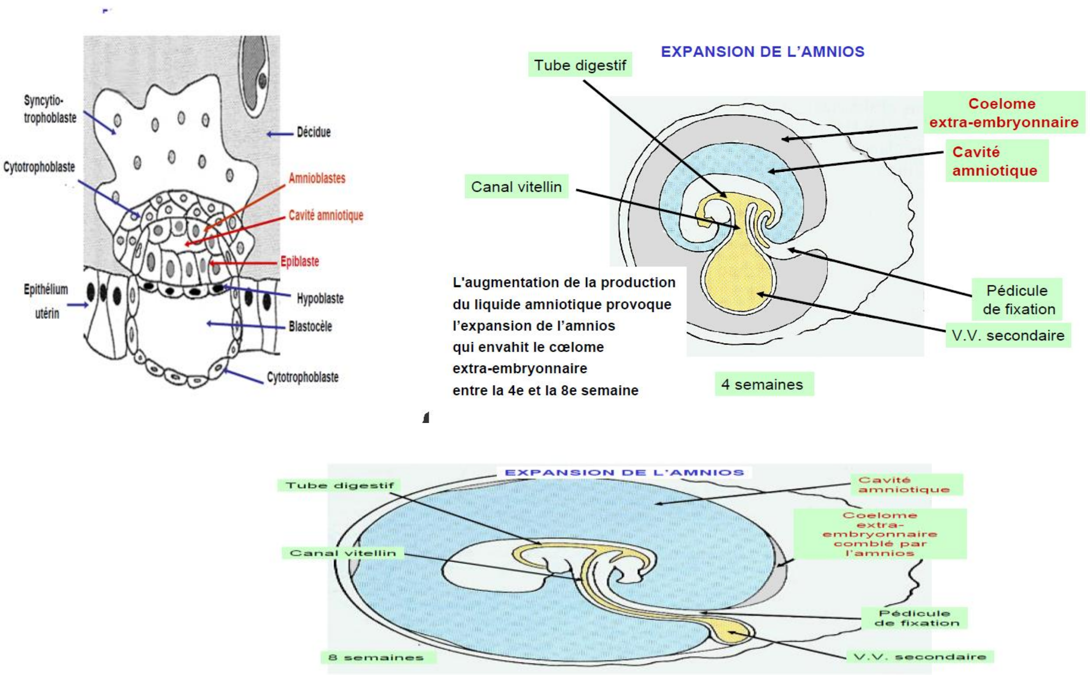


Figure 8. Expansion of the amnion and amniotic cavity

2.2. Umbilical or vitelline vesicle

The definitive yolk sac is a sac, located 'under' the belly of the embryo, whose wall is made up of the endoderm lined by the mesenchyme constituting the extra-embryonic splanchnopleure, communicating with the primitive digestive tract by the vitelline duct. Described as a yolk sac, it is delimited by a generally flattened epithelium lined with extra-embryonic mesenchyme.

At the end of the 3rd week, structures appear in the extra-embryonic mesenchyme, the Wolffian and Pander blood-vascular islets - hematopoietic until the 5th week - and the primordia of the umbilical vessels or Extra-embryonic mesenchyme includes the Wolffian and Pander blood-vascular islands—hematopoietic until the 5th week—and the primordia of the umbilical or vitelline vessels. The mesenchyme of the umbilical plate thus gives rise to the blood cell lineages whose development ensures hematopoiesis throughout the individual's life. From the point of delimitation, it is a very small structure, connected to the primitive gut by the umbilical or vitelline duct. Along with the vitelline duct, it is enclosed within the pedicle and then the umbilical cord, and its remnant is visible at the end of pregnancy at the point of insertion of the cord onto the placenta.

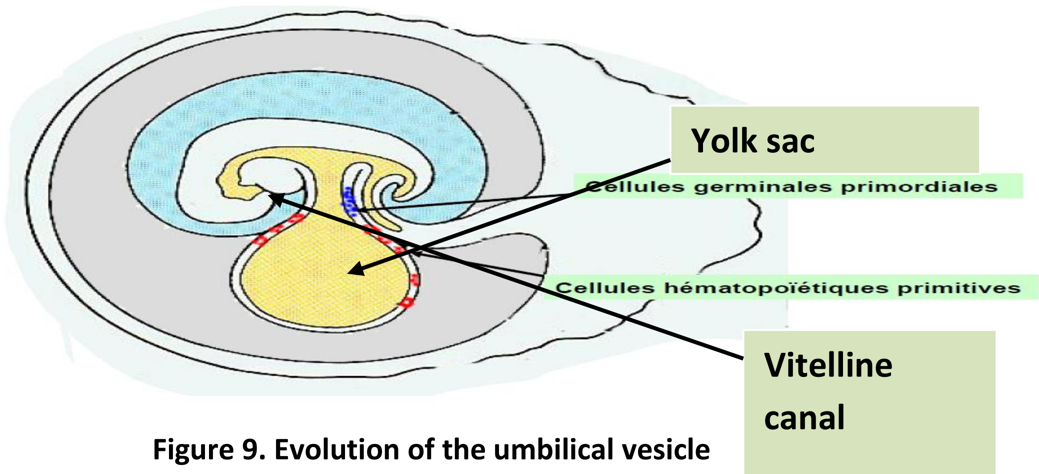


Figure 9. Evolution of the umbilical vesicle

2.3. Allantois

Defined as a diverticulum of the hindgut, this structure is visible in the caudal region of the embryo on day 17 of development. Its epithelium derives from the endoderm during the 3rd week and is lined by extra-embryonic mesenchyme. Within this mesenchyme, the primordial germ cells, which give rise to the germ line in each sex, are transiently located. The embryo's iliac vessels invade it to form the uterine system. Embryo-placental vascular. During the longitudinal delimitation of the embryo, the allantois is drawn into a ventral position, the posterior portion enters into the formation of the cloaca, then the urogenital sinus and contributes to forming the wall of the bladder.

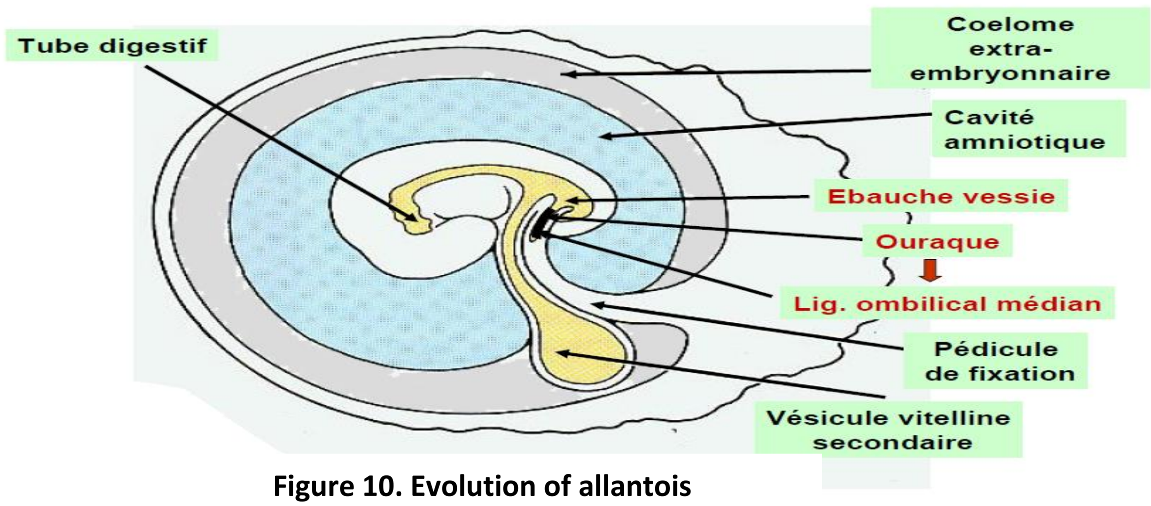


Figure 10. Evolution of allantois

2.4. Umbilical cord

It derives from the umbilical cord, which is formed at the end of the delimitation process. It lengthens as the pregnancy progresses and—at the end of pregnancy—measures 50 to 60 cm in length and 2 cm in diameter. It changes during pregnancy: its appearance in cross-section, in its middle portion, at the 4th month of pregnancy, can be used as a typical description. It is delimited by the amniotic epithelium, composed of extra-embryonic mesenchyme known as Wharton's jelly. Within the thickness of this formation, the following can be observed in cross-section: Two arteriesAllantoic or "umbilical" veins, the vitelline duct, the distal portion of the allantois. Thanks to the vessels it contains, it ensures the vascularization of the embryonic and then fetal portion of the placenta and consequently the exchanges with the maternal organism.

It can present length abnormalities which have harmful consequences at the time of delivery, and vascular abnormalities - presence of a single umbilical artery (1 case/200).

The role of the umbilical cord is to carry blood loaded with CO_2 and other waste products of fetal metabolism to the placenta via the umbilical arteries and to carry blood rich in O_2 and containing nutrients to the fetus via the umbilical vein.

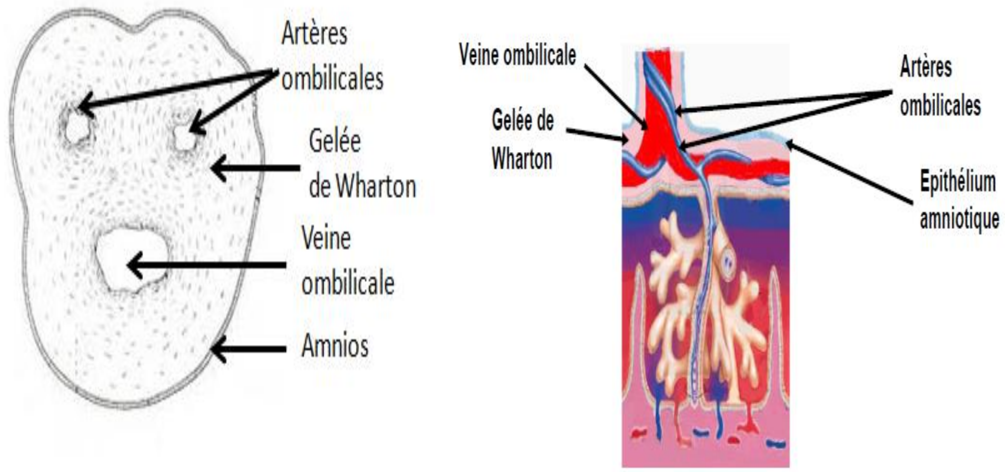


Figure 11. Structure of the umbilical cord at term